

Agent Concepts and MCP Basics

Explainer

An agent is distinct from an agentic workflow. A **workflow** is what most people use. **There is an LLM and we call it in a system that has predefined steps but these are determined deterministically via existing code or some AI workflow tool. An agent takes it one step further and is more autonomous. The agent chooses its processes and tools dynamically.** This means that it is given a task and it figures out the plan and how to execute it, what tools to use what functions to call without a human/code steering that process.

There are different types of agentic workflows which all hinge on the characteristic of an augmented LLM. In an agentic system the LLM is allowed to get information like query results, it's allowed to use tools and it's allowed to access memory. This give the LLM the correct information and context to be able to carry out complex tasks.

The first kind of workflow is **prompt chaining**. This is a workflow and not an agent because we have a set chain of sequencing. we call one LLM then feed the response of that into another LLM and segment the system into different tasks but we have a set step by step process which is the deterministic sequencing of the system.

The second kind of workflow is **routing**. This is is when we have a LLM that gets an input and then will decide between a set list of LLM's which to call based on the dynamic input. This is a workflow because we have a finite set of LLM options to call from and a predefined routing structure, even though the LLM dynamically decides which one to call based on the input.

The third is **parallelization workflow**. We have multiple LLM calls that run in simultaneously. This is a workflow because we have again, a finite set of LLM calls and predefined steps to get from input to output.

The fourth is **orchestrator workers** workflow. We have a central LLM that is segmenting the tasks and then it is delegating these to a number of LLMs and then will synthesize the result. This feels like an agent but it is not because the overall process is defined. One LLM Orchestrator will call one to many LLMs and then will synthesize the results. The number and kind of tasks the LLM orchestrator breaks the main task into can be unknown, but the orchestrator-workers pattern itself is predefined.

Finally we have the **evaluator optimizer**. This runs as a loop where one LLM creates the content which gets evaluated by another LLM and if the evaluator LLM is unhappy with the result it sends a response to the generator LLM which will create a new output and be

evaluated again by the other LLM and the loop goes until the evaluator is satisfied or max iterations are reached. This is a defined loop, still a workflow because we can outline the exact steps just not how many repeats.

Now an agent is different from the above workflows because we give the agent input and it is allowed to refer back to the human for judgement or more information but its sequencing of processes, how those tasks are carried out and with what tools happens independently. There is a trust needed in the reasoning ability of the LLM. I re-establish this distinction here because after introducing all the different types of workflows it is important to note that **an agent will establish its own workflows dynamically** it may set up a parallelization or routing workflow in its system after it has planned and reasoned about how to tackle the task. But how it will do so is up to its own judgement and dynamic and that is what draws the distinction between an agent and a workflow.

An **MCP** is precisely one of the ways that **we can augment an LLM and allow it to interact with the environment**. It allows an LLM/AI application/Chat interface to access resources (data that can be read from API/files), tools (functions that the LLM can call/execute with subject to permission of the AI/user) and prompts (previously engineered templates for different tasks). An MCP can, for example, allow Claude to send an email or read from an enterprise database or access the user's local files.

My **FL04 task was an automated company research and outreach preparation pipeline**. It consisted of 7 chained prompts, each step was predefined and set up by me. This means that it is a workflow since there was no agent dynamically planning and deciding on the steps to complete the task. In order for my FL04 task to become an agent it would need to be flexible and dynamic in choosing its processes and steps to completion. I would need to reword my prompt to show it the end result and goals of the system and **not to define the steps it needs to take to get there**. I could guide it to create subagents. It would be allowed to refer back to me for guidance if needed so I doesn't need to be fully autonomous but the main differentiation is that it may look like 7 sequential steps or it may implement some parallelization or orchestrator workflows dynamically as it reasons about how to achieve my shared goals and the required outcome and acceptance criteria.

(Word Count 896)

MCP Exploration

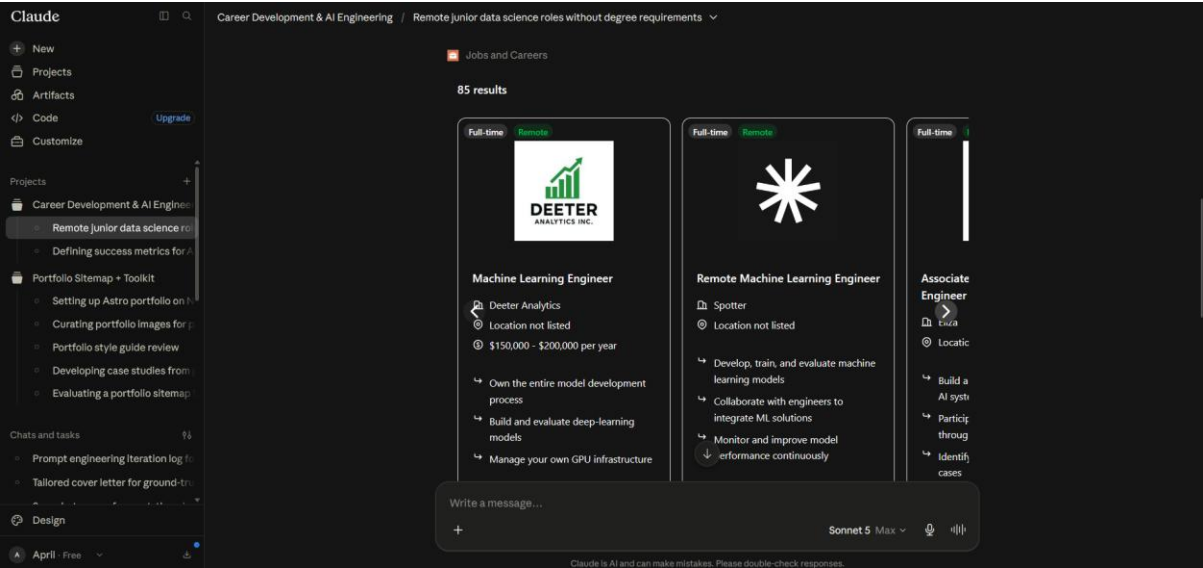
Tasks Outline

Task	What it's demonstrating
1. Discover	Query a live job database for relevant remote roles
2. Filter	Refine live results based on requirements/constraints
3. Experience-level search	Find a different subset internships, apprenticeships, and entry-level roles using live job data.

Task 1

Prompt: Find remote junior Data Scientist, Machine Learning Engineer, AI Engineer roles that are suitable for someone early in their career. Prioritize roles that do not explicitly require a bachelor's degree. Return the job title, company, location, salary if available, key requirements, and application route.

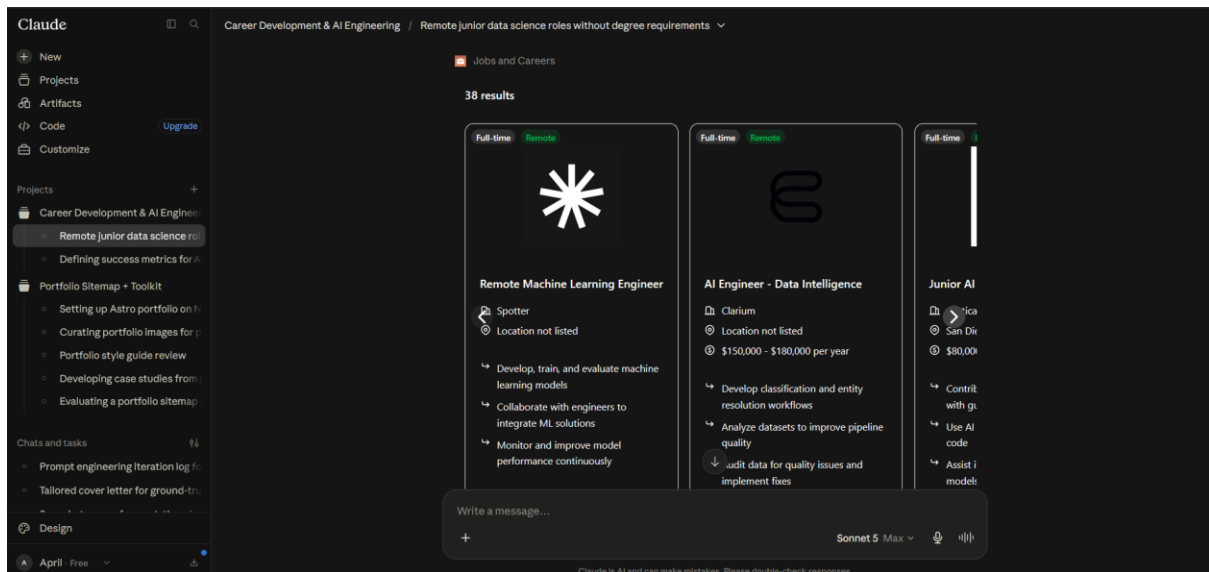
MCP call:



Task 2

Prompt: From those results, filter out any roles that explicitly require a bachelor's degree or more years of experience than I have. Prioritize fully remote roles and rank the remaining opportunities by how closely they match my Python, SQL, machine learning, and AI engineering experience.

MCP call:



Task 3

Prompt: Search for remote roles similar to the opportunities from my previous search, but specifically target internships, apprenticeships, and entry-level positions. Exclude roles requiring more than 2 years of experience.

MCP call:

